

Optimizing Data Center Siting in Texas Under SB6 Regulation

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1 Executive Summary

The rapid growth of artificial intelligence and cloud computing has increased demand for large-scale data center infrastructure. The International Energy Agency estimates that global data center electricity consumption reached approximately 415 TWh in 2024 and projects that it could more than double to roughly 945 TWh by 2030, growing far faster than overall electricity demand (International Energy Agency 2025). In Texas, this growth is especially important because ERCOT, Texas' electric grid operator, is facing a sharp increase in large-load interconnection requests, with state materials reporting that nearly 69% of 189 GW of large-load requests were associated with data centers (Public Utility Commission of Texas 2025). Texas Senate Bill 6 (SB6), enacted in 2025, requires the Public Utility Commission of Texas to establish interconnection standards for large loads in ERCOT, with a statutory demand threshold of 75 MW unless the Commission adopts a lower threshold (Texas Legislature 2025). Proposed implementation materials and legal summaries describe a \$50,000/MW non-refundable interconnection fee and study fees that vary by project size (Greenberg Traurig 2026; DLA Piper 2026). These rules change the economics of large-scale data center siting.

This report develops a tract-level optimization framework to evaluate where 10,000 MW of new data center capacity should be located in Texas under cost, environmental, land availability, accessibility, and SB6-related constraints. The model uses a merged dataset of 4,355 candidate census tracts combining ERCOT electricity prices, World Resources Institute water risk indicators, Bureau of Transportation Statistics noise data, U.S. Census population data, county-level land price estimates, and National Land Cover Database land availability measures. These inputs allow the model to compare candidate sites using a common geographic unit while accounting for differences in operating cost, community

exposure, hydrological risk, background noise, land availability, and distance to population centers.

After using preliminary LP and base MIP formulations to test the siting framework, the final analysis focuses on an SB6-aware mixed-integer linear model implemented in Gurobi. The model allocates capacity across census tracts while enforcing land, environmental, population exposure, and accessibility constraints. It also represents SB6 policy tiers by distinguishing sub-75 MW sites, SB6-triggering sites, and large hyperscale sites. Auxiliary variables are used to linearize the SB6 cost term so that the final formulation remains a mixed-integer linear program.

The results show that SB6-style thresholds change the spatial pattern of development. The optimal SB6-aware solution uses a small number of large sites and many facilities placed just below the 75 MW threshold. This indicates that interconnection rules with a sharp cost jump at 75 MW can create incentives for capacity fragmentation. Sensitivity analysis further shows that distance and population exposure constraints are among the most economically important policy levers, while moderate water and noise restrictions can often be satisfied with smaller cost increases. These curves also help identify Pareto-relevant parameter ranges, where policy goals can be strengthened without imposing steep marginal costs. A subsidy-efficiency analysis highlights census tracts where targeted incentives could shift development toward sites with better environmental performance at relatively low cost.

Based on these findings, we make four recommendations. First, the Public Utility Commission of Texas and ERCOT should replace the discontinuous 75 MW SB6 trigger with a smoother sliding-scale fee schedule to reduce artificial fragmentation. Second, the Texas Economic Development and Tourism Office (TEDO) should treat distance to population centers as an explicit incentive-design tradeoff, because stricter accessibility requirements increase system cost. Third, Texas planning agencies should use clear environmental thresholds for water stress, background noise, and population exposure rather than broad site rankings. Fourth, TEDO should screen incentive awards using subsidy-efficiency metrics so public dollars target tracts with measurable environmental benefits. Together, these recommendations show how the government of Texas can guide data center growth toward locations that are cost-efficient, environmentally feasible, and aligned with public infrastructure goals.

2 Introduction and Project Overview

2.1 Policy Context

Data centers are becoming a major source of electricity demand as artificial intelligence, cloud computing, and digital services expand. The International Energy Agency estimates that global data center electricity consumption was about 415 TWh in 2024 and projects that it could more than double to roughly 945 TWh by 2030 (International Energy Agency 2025). Lawrence Berkeley National Laboratory also finds that U.S. data center electricity

use has increased substantially and will remain a major power-sector planning issue (Shehabi et al. 2024).

Texas is especially important because ERCOT is facing rapid growth in large-load requests. Public Utility Commission of Texas materials from 2025 reported that ERCOT was tracking more than three times as many large-load interconnections as in 2024, with data centers representing nearly 69% of 189 GW of large-load requests (Public Utility Commission of Texas 2025). Senate Bill 6 was enacted in 2025 to address large-load interconnection standards, cost allocation, reliability, and stranded infrastructure risk in ERCOT (Texas Legislature 2025; Bracewell 2025). Proposed implementation summaries describe a 75 MW applicability threshold, study fees, and a \$50,000/MW non-refundable interconnection fee for large loads (Greenberg Traurig 2026; DLA Piper 2026).

Beyond grid interconnection, data centers also create local environmental and community impacts through electricity use, cooling-related water use, land requirements, and noise concerns (Mytton 2021; Shehabi et al. 2024). Since construction requires physically suitable land, we also use the National Land Cover Database, which measures U.S. land cover using satellite-derived 30-meter products (U.S. Geological Survey 2025; Multi-Resolution Land Characteristics Consortium 2024). These factors motivate a siting framework that jointly considers cost, regulation, land availability, environmental burden, and accessibility.

2.2 Decision Problem and Operations Research Framework

The central decision problem is: *how should a fixed amount of new data center capacity be allocated across Texas census tracts to minimize development and operating cost while satisfying regulatory, environmental, land availability, and accessibility constraints?*

This is a constrained spatial allocation problem. The decision-maker must choose where capacity is placed, how much each selected site receives, and whether selected sites trigger SB6-related cost tiers. Operations research is appropriate in this setting since linear and mixed-integer programming can represent site-level costs, capacity limits, environmental thresholds, binary site-selection decisions, and regulatory discontinuities in one model. Unlike a simple ranking approach, this framework guarantees feasibility across system-wide constraints and identifies which policy constraints are most economically important.

2.3 Contribution and Roadmap

This project makes four contributions:

- **Integrated tract-level dataset:** We merge 4,355 Texas census tracts with ERCOT electricity prices, water risk indicators, BTS noise data, Census population data, land price estimates, and USGS land cover data.

- **Optimization model with regulatory thresholds:** We formulate a mixed-integer linear model that allocates data center capacity while incorporating land availability, population exposure, water stress, background noise, accessibility, and SB6 threshold costs.
- **Sensitivity and shadow-price analysis:** We evaluate how total cost responds to changes in water, noise, population, and accessibility thresholds. This allows us to identify Pareto-relevant parameter ranges: thresholds where policy goals can be strengthened with limited cost increases, and thresholds where further tightening becomes economically inefficient. We also report conditional shadow prices after fixing the selected MIP structure.
- **Policy translation:** We translate the model results into recommendations for the Public Utility Commission of Texas, ERCOT, and the Texas Economic Development and Tourism Office.

The report proceeds as follows. Section 3 describes the data sources, cleaning, and joins. Section 4 presents exploratory data analysis. Section 5 defines the methodology. Section 6 presents experiments and results. Section 7 gives policy recommendations. Section 8 discusses limitations and future work. Section 9 provides references.

3 Data Sources and Initial Filtering

3.1 Raw Data Sources

The analysis combines six Texas-based sources into census-tract-level model inputs. Census tracts are used as the final unit of analysis because they allow population, environmental, land availability, and cost variables to be compared at a consistent spatial level.

Summary of the raw data sources used in the analysis.

Source	Organization	Raw structure	Main use in this project
Transportation noise	Bureau of Transportation Statistics	Spatial noise map	Proxy for existing local noise exposure
Water risk	World Resources Institute	Geospatial water-risk layer	Proxy for water stress and hydrological vulnerability
Electricity and fuel mix	Electric Reliability Council of Texas	Time-series market data by settlement point and load zone	Proxy for electricity operating cost
Population	U.S. Census Bureau	Census tract table	Measure of community exposure and demand

Source	Organization	Raw structure	Main use in this project
Rural land price	Texas Real Estate Research Center	County-level table from regional land market summaries	proximity Proxy for land acquisition cost
Land cover	U.S. Geological Survey	Raster land cover classes	Measures developable land availability and tract capacity limits

3.2 Filtering, Cleaning, and Joins

Each raw source is first converted into a tract-level input. Noise and land cover are summarized by tract, water risk is spatially joined to tract boundaries, ERCOT prices are assigned through load zones, population is already tract-level, and county land prices are mapped to all tracts within each county. This preprocessing ensures that variables from different geographic resolutions can be compared at the census-tract level.

After these transformations, the sources are merged into one tract-level dataset. Rows are kept only if they contain the required cost, environmental, land availability, and accessibility inputs needed for the optimization model. Table 2 summarizes each source's raw size, retained variables, and tract-level transformation.

Summary of raw dataset sizes, retained variables, and tract-level transformations after preprocessing.

Source	Shape	Retained variables	Tract-level transformation
Transportation noise	6,884 rows by 3 columns	tract identifier, avg_noise	Already summarized by tract
Water risk	229 rows by 17 columns	Water Stress, Water Depletion, Flood Risk, Drought Risk	Spatially joined to tracts
ERCOT electricity	70,080 rows by 6 columns	Settlement Point, Settlement Point Price	Averaged by load zone and assigned to tracts
Population	6,896 rows by 2 columns	tract identifier, total population	Already tract-level
Rural land price	248 rows by 3 columns	county name, price_per_acre	Assigned to tracts by county
Land cover	6,896 rows by 26 columns	tract identifier, dc_total_km2, dc_total_pct	Summarized by tract and converted to acres

3.3 Final Model-Ready Dataset

After filtering and joining, the final dataset contains 4,355 model-ready tracts out of 6,896 Texas census tracts. Removed tracts are excluded because at least one required model input could not be matched after the joins. Each remaining row represents one candidate census tract used in the optimization.

Figure 1 shows an excerpt of the final model-ready dataset. The tract code is recorded in TRACTCE, while NAMELSADCO records the county name. Variables such as price_per_acre and Settlement Point Price are included as tract-level assigned proxies, even though they originate from county-level and load-zone-level sources.

	TRACTCE	NAMELSADCO	Water Stress	Population	avg_noise	price_per_acre	Settlement Point Price	dc_total_acre	Population_Distance_miles
0	100	Victoria County	3.989606	1767	55.141090	6107.0	33.327444	1720.888641	197.172731
1	101	Lamar County	2.778739	4639	55.122428	9159.0	33.401317	4531.510332	257.643141
2	102	Angelina County	1.519398	4983	55.137782	9159.0	33.401317	11171.987708	193.070194
3	103	Bandera County	3.989606	3913	54.779465	7911.0	38.166779	11921.679567	208.910022
4	104	Wilson County	3.989606	4956	55.081008	6107.0	33.327444	15654.571249	190.308306
...	...	...	...	...	...	...	...	...	...
4350	980100	Bexar County	3.879872	1083	55.187314	7911.0	37.088334	1561.876573	199.114592
4351	980200	Jefferson County	1.158481	6169	55.137727	11502.0	34.566280	137.217407	225.044738
4352	980300	Harris County	2.522353	2849	55.098536	11502.0	34.566280	146.780370	171.074931
4353	980400	Harris County	2.522353	3751	55.150876	11502.0	34.566280	68.052717	172.597133
4354	980700	Harris County	2.522353	2299	55.150876	11502.0	34.566280	14.233248	171.041790

Excerpt of the final tract-level model-ready dataset used in the optimization.

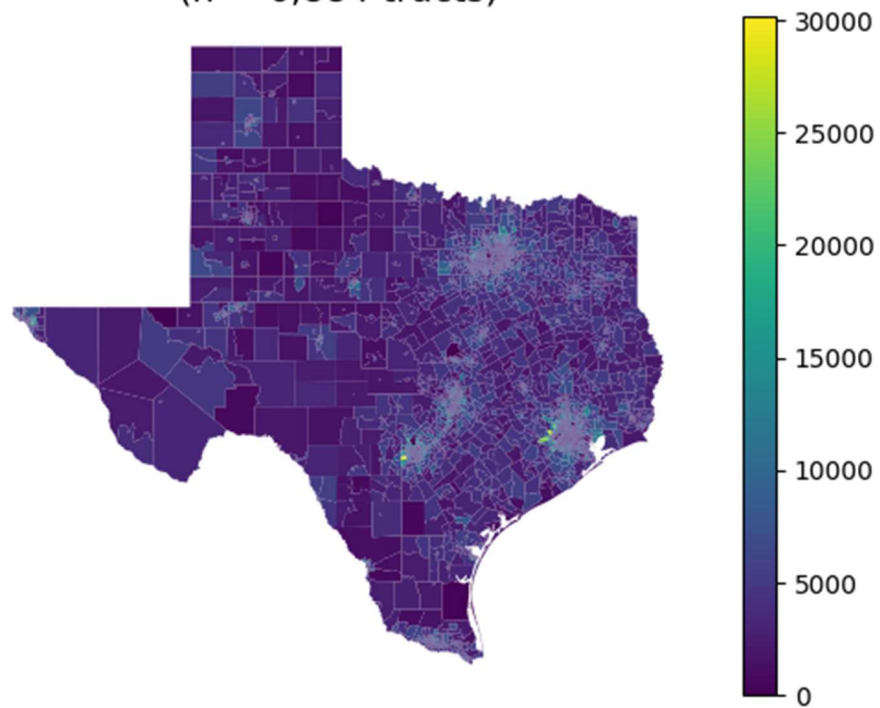
4 Exploratory Data Analysis

The exploratory analysis identifies the spatial patterns that motivate the optimization model. Texas is not a uniform siting environment: population, noise, water risk, electricity prices, land costs, and developable land vary across space. This means that a policy-relevant solution must evaluate several constraints together.

4.1 Population, Noise, and Water Patterns

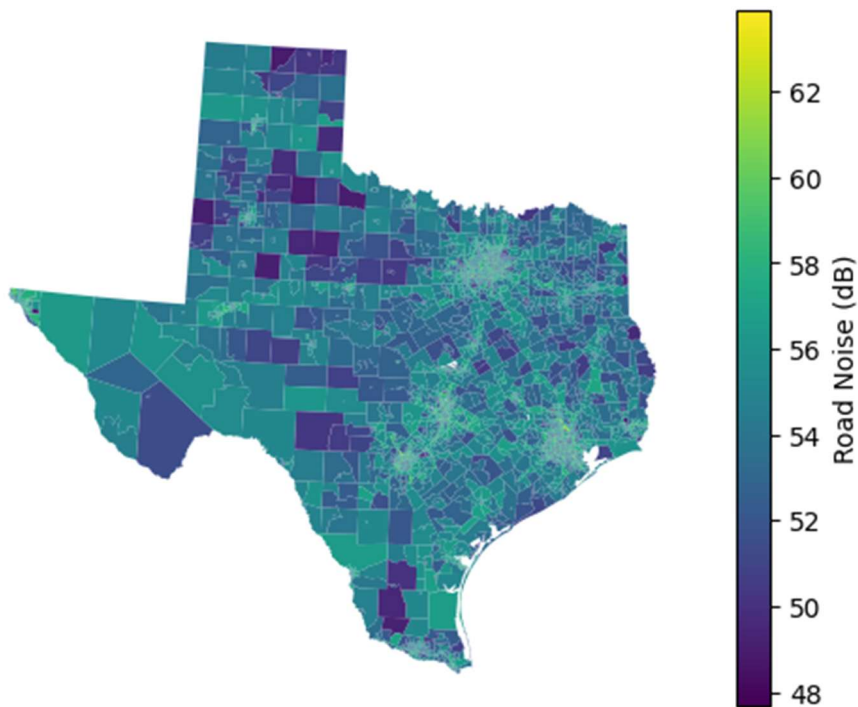
Figure 2 summarizes three spatial variables that directly shape the model: population exposure, transportation noise, and water stress. Population motivates the community exposure and accessibility constraints; noise provides a proxy for existing local infrastructure burden; and water stress captures hydrological risk relevant to data center cooling and long-run reliability (Mytton 2021).

Population in Texas by Census Tract
(n = 6,884 tracts)

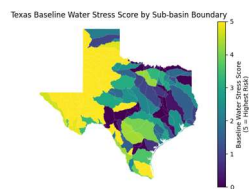


(a) Population by census tract

Road Noise in Texas by Census Tract (n = 6,884 tracts)



(b) Average road noise by tract



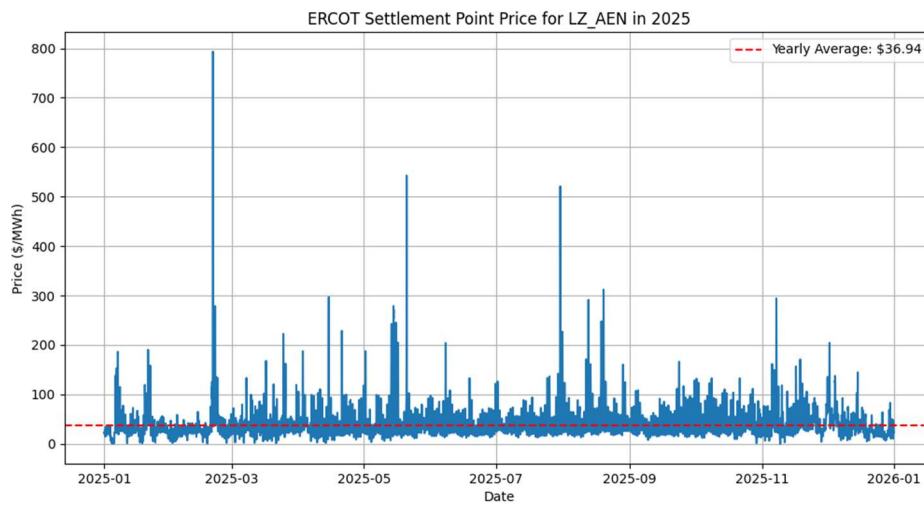
(c) Baseline water stress

Spatial patterns in population exposure, transportation noise, and water stress across Texas.

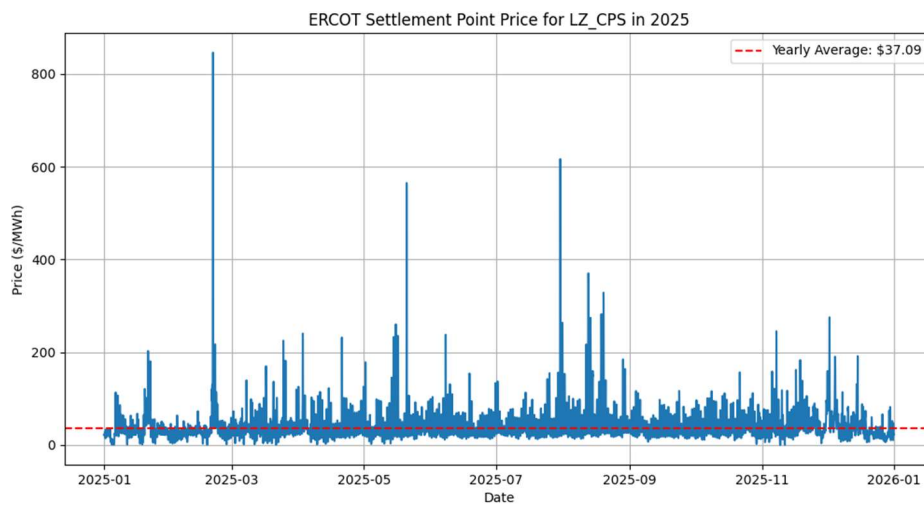
The maps show clear spatial tradeoffs. Population is concentrated around Dallas–Fort Worth, Houston, San Antonio, and Austin, while West Texas and the Panhandle are sparse. Noise is highest near major cities and transportation corridors, and water stress is generally higher in western Texas. Together, these patterns motivate the model’s population exposure, distance accessibility, background-noise, and water stress constraints.

4.2 ERCOT Electricity Price Conditions

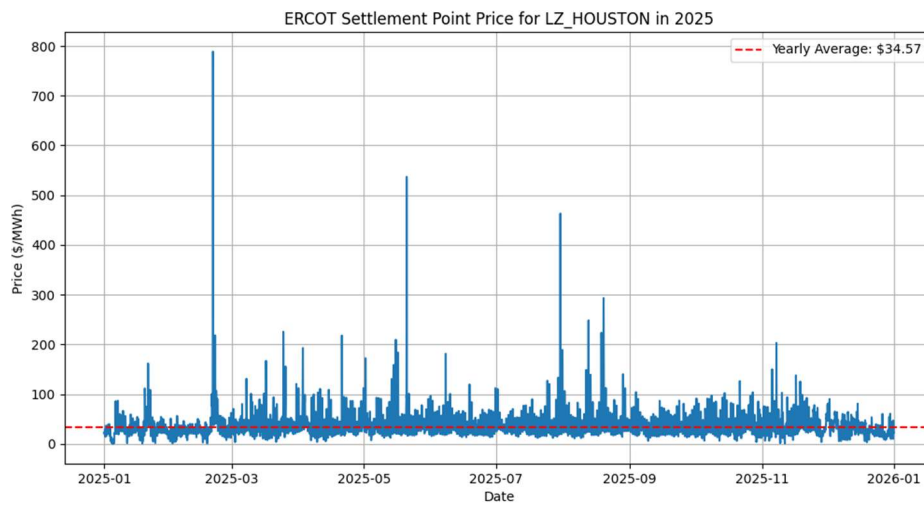
Data centers are electricity-intensive, making electricity cost and volatility central to siting decisions (Shehabi et al. 2024). Figure 3 shows 2025 ERCOT settlement point prices by load zone, highlighting variation in both average prices and price spikes.



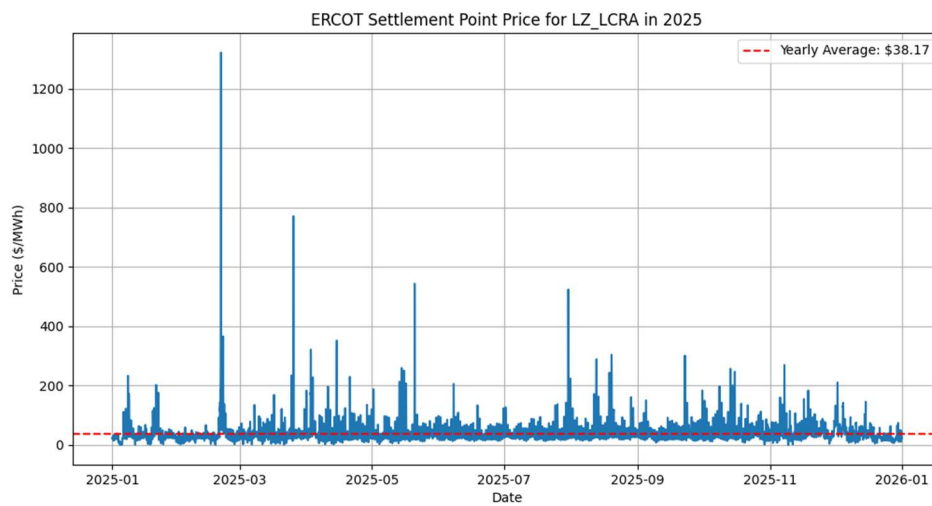
AEN



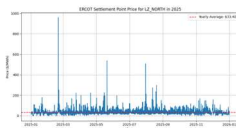
CPS



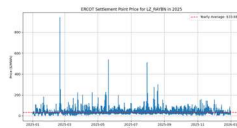
HOUSTON



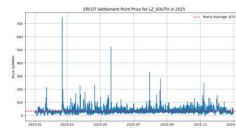
LCRA



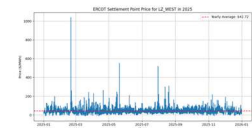
NORTH



RAYBN



SOUTH



WEST

Hourly ERCOT settlement point prices across load zones during 2025.

The exploratory data analysis supports using a load-zone electricity cost term instead of a statewide average.

5 Methodology

5.1 Modeling Overview

We model data center siting as a mixed-integer linear optimization problem. The model allocates fixed new capacity across Texas census tracts while minimizing cost and satisfying regulatory, environmental, land availability, and accessibility constraints. The formulation follows the exploratory data analysis: water stress, noise, population, electricity prices, land cover, and SB6 policy each enter as either objective terms or constraints.

5.2 Decision Variables

Let I be the set of candidate census tracts. The main decision variable is

$$x_i \geq 0 \quad \forall i \in I,$$

where x_i is the MW of data center capacity allocated to tract i . The binary variable

$$z_i \in \{0,1\} \quad \forall i \in I$$

equals 1 if tract i is selected. For the SB6-aware model, we add

$$s_i \in \{0,1\}, \quad h_i \in \{0,1\},$$

where $s_i = 1$ if site i triggers SB6 and $h_i = 1$ if it qualifies as a large hyperscale site. The tier hierarchy is

$$h_i \leq s_i \leq z_i \quad \forall i \in I.$$

5.3 Cost and Capacity Parameters

For each tract i , let l_i be land price per acre, c_i average electricity price, w_i water stress, n_i background noise, P_i population, d_i population-weighted distance, and A_i developable acres.

Assuming 1 MW requires 0.25 acres, tract capacity is

$$U_i = 4A_i.$$

The baseline cost per MW is

$$g_i = 0.25l_i + 8760c_i.$$

This combines land cost and one year of electricity exposure as a siting cost proxy, not a full lifecycle cost.

5.4 Base Capacity and Site Constraints

The model allocates fixed total capacity:

$$\sum_{i \in I} x_i = K.$$

The final SB6-aware run uses $K = 10,000$ MW, a 75 MW SB6 threshold, and a 250 MW large-site threshold.

Capacity must be assigned only to selected tracts, remain within land-cover capacity, and meet a minimum viable site size:

$$x_i \leq U_i z_i \quad \forall i \in I,$$

$$x_i \geq q_{\min} z_i \quad \forall i \in I.$$

5.5 Environmental and Accessibility Constraints

Because total capacity is fixed at K , average portfolio constraints remain linear. Water stress is limited by

$$\sum_{i \in I} w_i x_i \leq W^{\max} K.$$

Background noise is modeled as a minimum noise floor:

$$\sum_{i \in I} n_i x_i \geq N^{\min} K.$$

This favors areas with existing background noise while recognizing that already burdened communities should not receive unlimited additional burden.

Population exposure is capped by

$$\sum_{i \in I} P_i x_i \leq P^{\max} K,$$

and population proximity is enforced by

$$\sum_{i \in I} P_i x_i \geq P^{\min} K.$$

The first limits exposure in dense tracts; the second prevents all capacity from moving to remote areas. Distance accessibility is constrained by

$$\sum_{i \in I} d_i x_i \leq D^{\max} K.$$

5.6 SB6 Tier Modeling

SB6 applies to large-load interconnection standards in ERCOT, with 75 MW as the default statutory threshold unless a lower threshold is adopted (Texas Legislature 2025). Proposed

implementation summaries describe a \$50,000/MW non-refundable fee and size-based study fees (Greenberg Traurig 2026; DLA Piper 2026). We model three tiers: below 75 MW, at or above 75 MW, and at or above 250 MW.

To apply the per-MW fee only to SB6-triggering sites, define

$$y_i = s_i x_i.$$

Because this product is nonlinear, we use McCormick constraints:

$$y_i \leq U_i s_i,$$

$$y_i \leq x_i,$$

$$y_i \geq x_i - U_i(1 - s_i),$$

$$y_i \geq 0.$$

The SB6-aware objective is

$$\min \sum_{i \in I} g_i x_i + 50000 \sum_{i \in I} y_i + 100000 \sum_{i \in I} s_i + 200000 \sum_{i \in I} h_i.$$

This includes baseline siting cost, the SB6 per-MW fee, the base study fee, and the additional large-site study fee.

5.7 Model Development and Sensitivity

We first use simpler LP and base MIP formulations to test the structure of the siting model. These preliminary versions help verify that the cost terms, capacity limits, land availability constraints, and environmental constraints behave as expected before adding the full SB6 policy structure. The report's main analysis then focuses on the final SB6-aware MIP, because this is the model that directly represents the regulatory mechanism of interest: the 75 MW threshold, SB6-triggering sites, and large hyperscale sites.

After solving the SB6-aware MIP, we perform sensitivity analysis to understand how the solution changes when policy thresholds are tightened or relaxed. Because mixed-integer programs do not have standard global dual values, we fix the selected binary variables and re-solve the resulting LP to compute conditional shadow prices. These shadow prices measure local marginal tradeoffs around the selected site structure, so they should be interpreted as conditional sensitivity measures rather than global dual values for the original MIP.

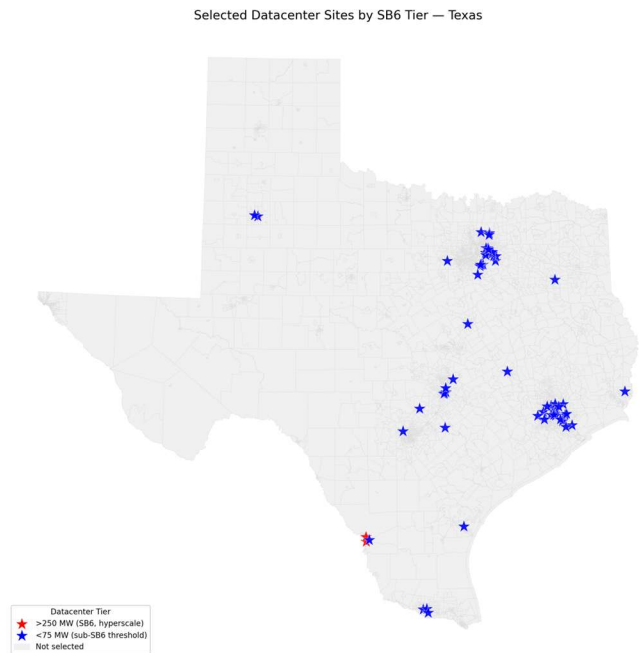
6 Experiments and Analysis

The experiments focus on the final SB6-aware MIP because this is the model that directly captures the regulatory mechanism motivating the project. The preliminary LP and base

MIP formulations were used to develop and test the model structure, but the policy analysis below is based on the final 10,000 MW SB6-aware specification.

6.1 Selected Sites and SB6 Tier Pattern

The SB6-aware model selects many sub-threshold sites (< 75 MW) and a small number of large sites. Figure 4 maps the selected sites by tier: blue markers indicate sites below 75 MW, while red markers indicate sites above 250 MW. This pattern is consistent with the incentive created by a discontinuous threshold.



Selected data center sites by SB6 tier. Blue stars are sub-75 MW sites and red stars are large sites above 250 MW.

Table 3 reports the ten largest selected sites. The largest selected sites are in Webb County, while many additional sites are set just below the 75 MW threshold. Values printed as 74.999 MW should be interpreted as just below 75 MW; the model uses this to avoid crossing the SB6 threshold.

Top ten selected sites in the SB6-aware solution. Sites listed as 74.999 MW are interpreted as just below the 75 MW threshold.

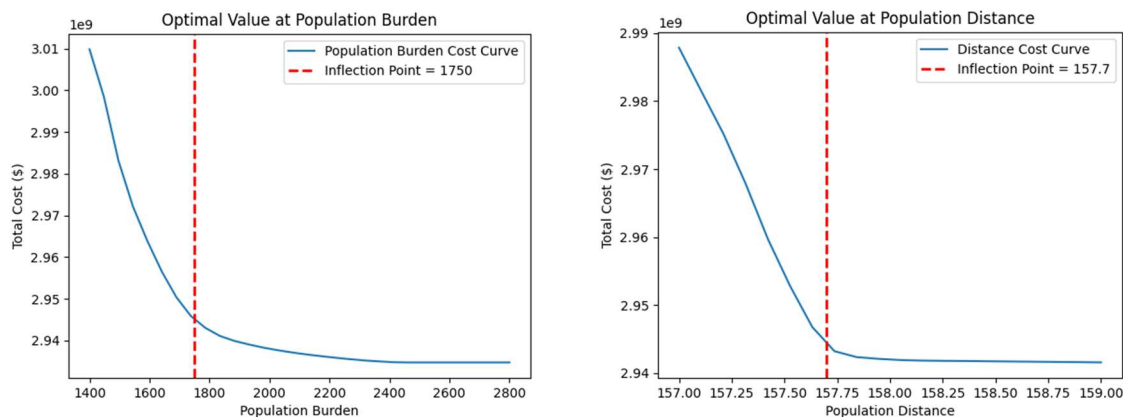
Model idx	County	MW	% of K	Tier	Base cost/MW	SB6 cost	Pop. dist.
87	Webb County	3848.52	38.49	> 250 MW	294573.29	192726000	152.93
168	Webb County	620.03	6.20	> 250 MW	294573.29	31301740	153.10

Model idx	County	MW	% of K	Tier	Base cost/MW	SB6 cost	Pop. dist.
225	Nueces County	74.999	0.75	< 75 MW	294573.29	0	152.93
4239	Harris County	74.999	0.75	< 75 MW	304778.36	0	160.41
4329	Harris County	74.999	0.75	< 75 MW	294573.29	0	152.23
4341	Harris County	74.999	0.75	< 75 MW	294573.29	0	152.85
303	Lubbock County	74.999	0.75	< 75 MW	325578.38	0	164.35
1157	Dallas County	74.999	0.75	< 75 MW	293926.16	0	158.44
1161	Dallas County	74.999	0.75	< 75 MW	293926.16	0	159.54
1165	Dallas County	74.999	0.75	< 75 MW	293926.16	0	159.17

The main result is that the threshold creates a mixed portfolio: a small number of large anchor sites plus many sub-threshold sites. This does not necessarily mean the threshold is bad, but it shows that the policy changes the spatial incentives facing developers.

6.2 Sensitivity Curves

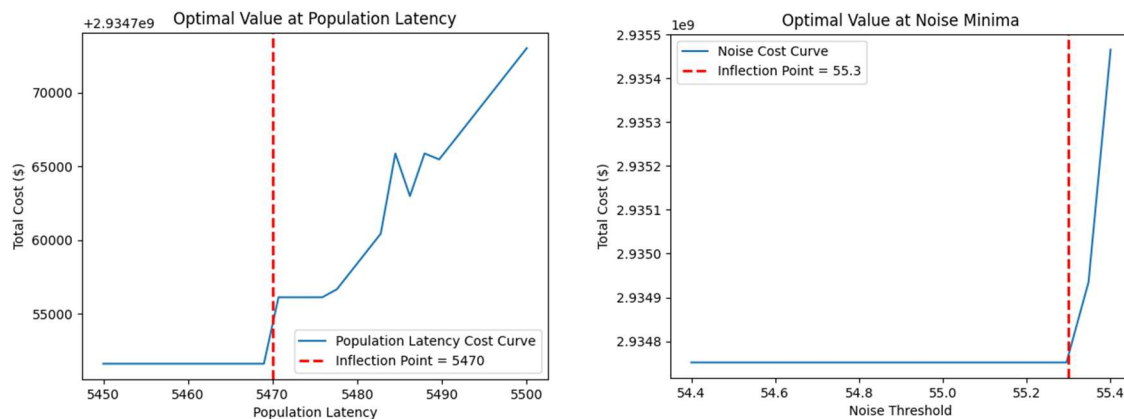
Figures [fig:burden_curve]–7 show how total cost changes as policy thresholds vary. The purpose of these curves is to identify threshold ranges that are Pareto-relevant in practice: points where environmental or accessibility goals improve without large cost increases, and points where stricter requirements begin to impose steep marginal costs. In this sense, the sensitivity analysis helps distinguish efficient policy parameter choices from thresholds that are overly restrictive relative to their cost.



Total cost as the population-weighted distance threshold is relaxed.

The population burden curve declines sharply at low thresholds and then flattens around a burden level of 1,750. This means strict burden limits force the model into recommending more expensive locations, while further relaxing the constraint beyond the inflection point would yield diminishing savings.

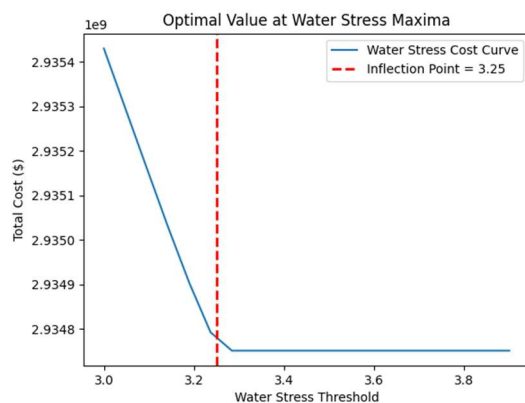
The distance curve shows that proximity to population centers is costly. It flattens around 157.7 miles, indicating that once the model has enough geographic flexibility, additional relaxation does not substantially reduce cost.



Total cost as the minimum background-noise threshold increases.

The population latency curve rises after the threshold passes about 5,470. This suggests that stronger proximity requirements become costly once the model is forced toward more demand-adjacent locations.

The noise curve is flat for moderate thresholds but rises sharply after about 55.3. This means the model can satisfy a moderate background-noise preference without much cost, but requiring all selected capacity to be in the noisiest tracts becomes expensive.



Total cost as the maximum water stress threshold is relaxed.

The water stress curve declines until roughly 3.25 and then flattens. This suggests that very strict water stress constraints increase cost, but moderate water protection can be imposed with limited additional cost.

Overall, the sensitivity curves show that the key policy choice is not simply whether to impose constraints, but where to set their thresholds. The flatter regions of the curves identify Pareto-relevant parameter ranges where environmental and accessibility requirements can guide siting with limited cost increases, while the steep regions indicate thresholds where additional strictness becomes economically inefficient.

6.3 Conditional Shadow Prices

Table 4 reports conditional shadow prices after fixing the MIP-selected site structure and re-solving the resulting LP. These values complement the sensitivity curves by measuring local marginal tradeoffs around the selected solution. The sign of each value follows the constraint convention used by Gurobi, so interpretation should focus on the direction of the constraint and the relative magnitude.

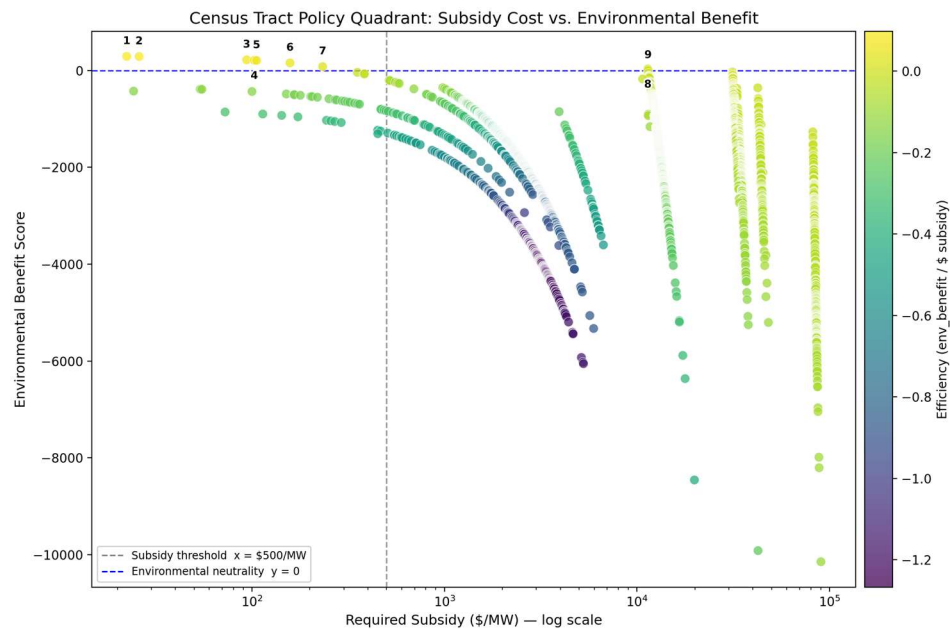
Conditional shadow prices after fixing the MIP site-selection structure, ordered by absolute magnitude.

Constraint	Shadow price	Interpretation
Total capacity	-793025.70	Conditional shadow price for increasing required capacity by 1 MW
Noise preference	27746.08	Cost effect of changing the minimum noise floor, scaled by total capacity
Distance accessibility	-2274.32	Value of relaxing the distance cap, scaled by total capacity
Water stress	-111.60	Value of relaxing the water stress threshold, scaled by total capacity
Population burden	-29.56	Value of relaxing the population burden cap, scaled by total capacity
Population latency	0.00	No marginal effect at the selected solution

The largest economically meaningful values are associated with total capacity, distance accessibility, and the noise preference. Population latency has zero marginal value at the selected solution, meaning that it is not the active marginal constraint at that point. The signs should be read relative to how each constraint is written: relaxing an upper-bound constraint generally lowers cost, while tightening a lower-bound constraint can increase cost.

6.4 Subsidy Efficiency Analysis

To further extend the analysis, we examine whether targeted subsidies could shift development toward environmentally preferable tracts not selected in the optimal SB6-aware solution. For each unselected tract, we estimate the required subsidy per MW using its reduced cost, or how much its effective objective value must improve before selection. We then compare this subsidy with the tract's environmental benefit, defined by its improvement relative to the model's environmental constraint bounds. Figure 8 plots this tradeoff. The dashed lines mark environmental neutrality and the \$500/MW subsidy threshold, making the upper left-hand quadrant the strongest target area: positive environmental benefit at relatively low subsidy.



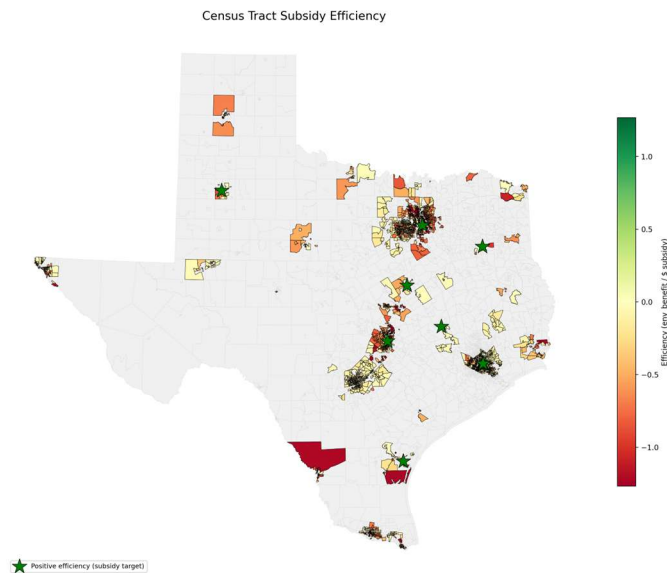
Census tract policy quadrant comparing required subsidy per MW with environmental benefit.

Top subsidy target tracts ranked by subsidy efficiency.

#	County	Tract	Required subsidy/MW	Environmental benefit	Efficiency
1	Smith County	2003	22.34	289.49	0.0983
2	Travis County	1504	25.91	285.92	0.0971
3	Nueces County	2001	93.80	218.03	0.0740
4	Dallas County	1503	102.80	209.03	0.0710
5	Brazos County	203	105.62	206.21	0.0700
6	Lubbock County	1601	157.54	154.29	0.0524
7	McLennan	2501	232.82	79.02	0.0268

#	County	Tract	Required subsidy/MW	Environmental benefit	Efficiency
	County				
8	Harris County	511502	11384.77	29.89	0.0026
9	Harris County	511400	11413.84	0.81	0.0001

Table 5 shows that the strongest subsidy candidates are unselected tracts where a relatively small subsidy could produce a meaningful environmental improvement. Smith County Tract 2003 and Travis County Tract 1504 provide the highest subsidy efficiency because they combine high environmental benefit with low required subsidies per MW. By contrast, the Harris County tracts at the bottom of the table require subsidies above \$11,000/MW but produce much smaller environmental benefits. This difference matters for policy because incentives should not be awarded broadly; they should target tracts where a small reduction in effective cost can shift the solution toward better environmental performance.



Map of subsidy efficiency across Texas. Green stars indicate positive-efficiency subsidy targets.

Figure 9 adds a spatial interpretation to the subsidy results. The subsidy-efficient tracts are not concentrated in one county or one metropolitan region; instead, they are distributed across multiple parts of Texas. This supports a portfolio-based incentive strategy. Rather than choosing one target geography, TEDO could prioritize several high-efficiency sites across the state, reducing geographic concentration while still directing public funds toward tracts where subsidies produce meaningful environmental gains.

6.5 Main Findings

The experiments support four main conclusions. First, SB6-style thresholds can change developer behavior by encouraging projects to stay just below the 75 MW cutoff. The selected-site results show a mixed portfolio with a small number of large anchor sites and many sub-threshold sites, suggesting that the regulatory cliff affects not only total cost but also where and how capacity is distributed. Second, distance and population exposure constraints have a meaningful cost impact. This means that access to demand centers and limits on community exposure are central siting tradeoffs, not secondary issues to consider after locations have already been chosen.

Third, the sensitivity curves show that environmental and accessibility constraints do not affect cost in a straight-line way. Moderate water, noise, and distance requirements can guide siting with limited cost increases, but very strict thresholds become expensive once the model runs out of lower-cost feasible alternatives. This helps identify practical parameter ranges where policy goals can improve without large marginal cost increases. Fourth, the subsidy analysis shows that targeted incentives can improve the environmental profile of the siting portfolio more efficiently than broad subsidies. The strongest policy lesson is that Texas should treat data center siting as a portfolio problem: regulatory design, environmental thresholds, accessibility, and subsidy targeting should be evaluated together rather than separately.

7 Policy Recommendations

The recommendations below are directed to Texas agencies with authority over interconnection policy, incentive design, and environmental review. They are scoped to the strongest model results: SB6 thresholds, accessibility requirements, environmental constraints, and subsidy design all affect optimal siting outcomes.

7.1 Recommendations

Recommendation 1: Replace the 75 MW SB6 Cliff with a Sliding-Scale Fee. (*Decision-maker: Public Utility Commission of Texas and ERCOT*). The Public Utility Commission of Texas and ERCOT should replace the hard 75 MW SB6 trigger with a sliding-scale interconnection fee. The SB6-aware model selects several sites at 74.999 MW, showing that the threshold encourages fragmentation. Rather than moving the limit, ERCOT could charge a lower fee for 50–75 MW projects, a higher fee for 75–250 MW projects, and the full fee above 250 MW. This would recover grid-planning costs while reducing incentives to split projects only to avoid the threshold.

Recommendation 2: Make Distance to Population Centers an Explicit Incentive Criterion. (*Decision-maker: Texas Economic Development and Tourism Office*). TEDO should include distance to population centers in data center incentive review. The sensitivity analysis shows that stricter distance requirements increase total cost, with the

distance curve flattening around 157.7 miles. This means proximity to demand is valuable but costly, so incentives for closer-in sites should be awarded only when the accessibility benefit justifies the added cost.

Recommendation 3: Adopt Clear Environmental Screening Thresholds. (*Decision-maker: Public Utility Commission of Texas and regional planning authorities*). The Public Utility Commission of Texas and regional planning authorities should use measurable environmental screens for large data center proposals. The model treats water stress, background noise, and population exposure as explicit constraints. For example, the water stress curve flattens around 3.25, while the noise curve stays mostly flat until about 55.3. Regulatory agencies can use these thresholds to set transparent limits before projects move forward.

Recommendation 4: Target Incentives Using Subsidy Efficiency. (*Decision-maker: Texas Economic Development and Tourism Office*). TEDO should rank incentive candidates by subsidy efficiency: environmental benefit per dollar of required subsidy. For instance, Smith County (Tract 2003) and Travis County (Tract 1504) have the highest efficiency scores, while some Harris County tracts require subsidies above \$11,000 per MW for much smaller environmental benefits. TEDO should prioritize high-efficiency tracts instead of offering broad incentives to projects likely to be built without public support.

7.2 Summary of Decision Implications

Together, these recommendations translate the model into specific actions. The Public Utility Commission of Texas and ERCOT should smooth SB6 fees; TEDO should design incentives around distance and subsidy efficiency; and Texas planning agencies should apply measurable environmental thresholds. Texas should evaluate data center proposals as part of a constrained infrastructure portfolio, not only by private development cost.

8 Limitations and Future Work

This analysis has several important limitations. The model relies on proxies: population-weighted distance does not fully capture latency, fiber access, or cloud demand; land cover does not confirm parcel ownership, zoning approval, floodplain suitability, environmental restrictions, or local acceptance; and county-level land prices may differ from parcel-level acquisition costs. Average ERCOT prices also omit congestion, interconnection capacity, demand charges, volatility, and firm power procurement costs. Water variables are regional indicators rather than facility-specific water rights or cooling technology choices, which can materially change water and electricity tradeoffs (Mytton 2021; Microsoft 2025).

Future work should incorporate market and parcel-level data, including fiber routes, substations, interconnection capacity, zoning, water rights, cooling technologies, and electricity price scenarios. The model could also be extended to stochastic or robust optimization to capture uncertainty in electricity prices, water availability, interconnection

costs, and demand growth. Finally, the noise constraint should be refined with environmental justice measures, and SB6 assumptions should be updated if final rules from the Public Utility Commission of Texas differ from the structures used in this report (Texas Legislature 2025; Greenberg Traurig 2026; DLA Piper 2026).

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